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A viable controlled splitting strategy to improve transmission systems resilience against blackouts

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ABSTRACT

An effective strategy for intentional islanding of a transmission system is proposed in this paper to boost its resilience under low-probability high-impact events such as man-made attacks and natural disasters, which can cause catastrophic blackouts. This strategy considers two main aspects of islanding, i.e. time and scheme. To realize the appropriate time that the system need to be intentionally islanded, a new algorithm based on analysis of the rotor angles of its generators is first presented to distinguish coherent groups of the generators. The main advantage of this algorithm is its independence to the predefined degree of coherency and number of the groups. The information obtained by grouping the generators is then used to compute a new defined indicator, whose sign change specifies the islanding time. To decide about the lines that must be disconnected for fulfillment of the islanding scheme, the initial islands are first formed by tracing the active power produced by the generators. Then, the intended scheme is devised by solving a binary linear programming (BLP) problem that minimizes the disruption of the power transferred between these islands. Comparing the proposed scheme with some existing ones shows that the proposed strategy splits two IEEE test systems into stable islands with a better voltage profile and lower imbalance between the active power generation and demand by disconnecting fewer lines, which have smaller amounts of the power flows.

1. Introduction

Disruptive events such as windstorm, flood, earthquake and attack to critical infrastructures can trigger cascading failures, which may leave millions of people without electric power and cause the damages priced in billions of dollars by leading to large blackouts [1–3]. Occurring many blackouts with great intensity around the world in recent decades due to such extraordinary events makes developing approaches for the resilience enhancement of the power systems against them to be vitally important. These approaches can be categorized in the existing literature as two main groups, i.e. hardening and operational [3,4]. Hardening measures, which include reinforcing the power grid components with more robust materials, elevating substations, building redundant transmission lines, vegetation management and so forth, are generally long-term and costly strategies that may not be enough to make the system capable of coping with the emergency conditions during extreme events [2–4]. As opposed to this type of resilience enhancement remedies, operational measures preserve reliable function of the power system in a cost-effective manner by applying real-time monitoring data to mitigate or restrict the impact of severe events [3,4].

Intentional islanding, which is the focus of this paper and refers to partitioning the power system into multiple stable and self-sufficient islands in a controllable way to isolate faulty areas, is an example of operational actions often used as the last resort to stop spreading cascading outages [3,4]. Each strategy presented in this field must answer two questions: (a) when and (b) where islanding should be carried out to inhibit system instability. However, a few approaches have been proposed so far to solve both problems simultaneously, e.g. [5–8].

A majority of the researchers work on designing the islanding scheme using various tools such as ordered binary decision diagrams [8–10], searching process [5,6,11], spectral clustering [1,3,7,12–16], genetic algorithm [17,18], ant search [19,20], binary imperialist competitive [21], flow tracing [22], fuzzy systems [7,23,24], proximity analysis [25], artificial neural networks (ANN) [26,27] and mixed integer linear programming [28–31]. Most of these schemes consider the coherency constraint of the generators in individual islands [1,5–15,18–23,30,31]. Nevertheless, the method used to identify the coherent generators in some of them is not discussed [13,14,18–20] or adequate explanation is not provided [1,12,22]. In some others, coherent generators are grouped either by executing time-domain

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simulations for different loading and contingencies to obtain the generators angular velocity or rotor angle curves and assuming an arbitrary value as the number of coherent groups [7,9,10,23], or by offline analysis of the system models like slow coherency [8,11,21,30,31]. The identification of coherent groups using the slow coherency method is independent of the size, type, and location of the disturbance, and somewhat independent of the system operating point. Nonetheless, changing the operating conditions or topology of a system can alter the coherent groups, i.e. a group may be split into smaller ones or conversely, multiple groups may form a larger one. The shortcoming of the method deployed in Ref. [5] is that the number of the groups depends on the threshold selected for the degree of coherency.

On the other hand, in the researches, which deal with determining the proper islanding time, ANN [5], support vector machine (SVM) [6], a combination of ANN and fuzzy systems [7], decision tree (DT) [32] and comparing probable separation risk index [8] or the value and change rate of the system frequency [33] with thresholds are employed to address this issue. Since training ANN and SVM, designing fuzzy systems and DT, and specifying thresholds need several simulations of possible severe contingencies, and these processes must be also repeated in the case of any change in the generation units or transmission equipment or production and consumption values of the system, these methods, whose appropriate function depends on fine-tuning various parameters, cannot be effectively implemented on all power systems.

This paper proposes a novel strategy for controlled splitting of a power system into stable islands to prevent blackout. In this strategy, coherent groups of the generators are identified by an online method, in which neither choosing the threshold for the degree of coherency nor executing several offline simulations for finding the proper number of the coherent groups is required. A new indicator, whose sign change exhibits when the system needs to be islanded, is introduced. This indicator is not case sensitive and could be used in any power system. Thus, it has no need to be compared with a predefined threshold, which must be obtained for every system based on its operating conditions. The active loss allocation by the power tracing based approach presented in Ref. [34] is then used for determining the domains of the generators that are in the same group to form the initial islands. To acquire final islanding scheme, a BLP problem is formulated and solved with the objective of minimizing the disruption of the total apparent power transferred between the initial islands.

The rest of this paper is organized as follows: the procedure of implementing two major parts of the proposed strategy is described comprehensively in Sections 2 and 3. Intentional islanding results of 39- and 118-bus IEEE test systems, which show bus voltages and frequencies of the stable islands are closer to the base values by applying the proposed strategy in comparison with some earlier ones, are presented in Section 4. At last, this paper concludes in Section 5.

2. Time of islanding

2.1. Identifying coherent generator groups

To identify coherent generator groups at each sliding time window, the symmetric matrix called generator similarity, i.e. $\mathbf{D_g}$, whose dimension is equal to the total number of generators, i.e. N_g, must be first formed. The entry of the γth row and the ζth column of this matrix, i.e. $[\mathbf{D_g}]_{\gamma,\zeta}$, which is computed using Eq. (1), represents the degree of coherency between generators γ and ζ based on a simple criterion denoted as the electromechanical coherency in Ref. [35]. Because if the rotor angle variations of these two generators in the study time window are completely similar to each other, $[\mathbf{D_g}]_{\gamma,\zeta}$ will equal to one, and if their difference increases, $[\mathbf{D_g}]_{\gamma,\zeta}$ will become smaller.

$$[\mathbf{D_g}]_{\gamma,\zeta} = [\mathbf{D_g}]_{\zeta,\gamma} = \begin{cases} 1 - \dfrac{max(\text{abs}(\delta_\gamma - \delta_\zeta)) - min(\text{abs}(\delta_\gamma - \delta_\zeta))}{\delta_{max}} & \text{if} \gamma \neq \zeta \\ 1 & \text{if} \gamma = \zeta \end{cases}$$
$$\text{for} \gamma = 1, 2, ..., N_g \ , \ \zeta = \gamma, \gamma + 1, ..., N_g \quad (1)$$

where *max*(.) and *min*(.) are the calculation operators that return the largest and smallest element (member) of a vector (set), respectively; abs(.) is the calculation operator, whose output is equal to the absolute value of its input; δ_γ and δ_ζ are N_s-dimension vectors of the rotor angle of generators γ and ζ in degree, respectively, which are sampled with a sampling rate of f_s in the sliding time window with a width of T_w; and thus $N_s = f_s \times T_w$; δ_{max} is the maximum root mean square (RMS) value of the rotor angle of the generators within the study time window. It should be noted that T_w must be small enough to correctly identify coherent generator groups without need to store a huge amount of data; and it must be large enough to properly analyze system behavior. Also, f_s must be twice the frequency of the study power system according to Ref. [36].

It is worth noting that the rotor angle of a given generator can be estimated from the amplitude and phase angle of the terminal voltage and the output current of that generator [37] or replaced by the phase angle of the generator terminal voltage due to strong correlation between these two angles [38]. Fortunately, all these electrical variables can be directly measured via phasor measurement units (PMUs) and their precise synchronized values are available in real time through the wide area measurement system (WAMS), if the power system is fully observable [37,38]. Hence, one of several methods, e.g. [39–41], can be used to determine the optimal locations of PMUs for maintaining full observability of the study system during multiple PMU failures or line outages.

The second step to acquire different groups of coherent generators is the insertion of $\mathbf{D_g}$ into the algorithm known as community detection. This algorithm takes $\mathbf{D_g}$ as the adjacency matrix of a weighted undirected graph and divides this graph into two subgraphs if it is dividable. The algorithm divides each subgraph into two new ones if a good division exists. The process of dividing will continue until the measure called modularity does not increase. This case happens when the modularity matrix of the subgraph, which is created using its adjacency matrix, has no positive eigenvalues. Further details of this algorithm can be found in Ref. [15]. It gives the vector as an output, the number of the elements of which is equal to the dimension of $\mathbf{D_g}$ and the value of each element indicates the group number of its corresponding vertex. Therefore, this vector specifies how many coherent groups are formed within the study time window and each generator belongs to which group. This eliminates the requirement for determining the number of coherent groups or the degree of coherency based on several simulations. The information provided by these two steps will be used in the following section to recognize the necessity of intentional islanding by calculating the system separation indicator, i.e. *SSI*.

2.2. Calculating system separation indicator

To compute *SSI* within a sliding time window, an equivalent rotor angle of each coherent group, e.g. $\delta_{\mathbf{C}_\eta}$ for group η, must be first evaluated using Eq. (2) based on the concept of center of inertia (COI) [5,7,23,38]:

$$\delta_{\mathbf{C}_\eta} = \frac{1}{\sum_{\gamma \in \mathbf{C}_\eta} H_\gamma} \sum_{\gamma \in \mathbf{C}_\eta} H_\gamma \times \delta_\gamma \quad \text{for} \eta = 1, 2, ..., N_c \quad (2)$$

where $\mathbf{C}_\eta$ is the ηth group of coherent generators and H_γ is the inertia constant of generator γ in second. Then the symmetric matrix called group similarity, i.e. $\mathbf{D_c}$, whose dimension equals the total number of

groups of coherent generators, i.e. N_c, must be constructed as follows:

$$[\mathbf{D_c}]_{\eta,\lambda} = [\mathbf{D_c}]_{\lambda,\eta} = \begin{cases} 1 - \dfrac{max(\text{abs}(\boldsymbol{\delta}_{\mathbf{C}_\eta} - \boldsymbol{\delta}_{\mathbf{C}_\lambda})) - min(\text{abs}(\boldsymbol{\delta}_{\mathbf{C}_\eta} - \boldsymbol{\delta}_{\mathbf{C}_\lambda}))}{\delta^{C}_{max}} & \text{if} \eta \neq \lambda \\ 1 & \text{if} \eta = \lambda \text{and } \text{size}(\mathbf{C}_\eta) = 1 \\ \dfrac{\sum_{\gamma \in \mathbf{C}_\eta} \sum_{\zeta \in \mathbf{C}_\eta, \zeta \neq \gamma} [\mathbf{D_g}]_{\gamma,\zeta}}{\text{size}(\mathbf{C}_\eta) \times (\text{size}(\mathbf{C}_\eta) - 1)} & \text{if} \eta = \lambda \text{and } \text{size}(\mathbf{C}_\eta) \neq 1 \end{cases}$$

$$\text{for} \eta = 1, 2, ..., N_c \ , \ \lambda = \eta, \eta + 1, ..., N_c \tag{3}$$

where δ^{C}_{max} is the maximum RMS value of the equivalent rotor angle of the groups within the study time window and size(.) is the calculation operator that returns total number of members of a set. In fact, $[\mathbf{D_c}]_{\eta,\lambda}$, which is the entry of the ηth row and the λth column of $\mathbf{D_c}$, represents the degree of coherency between two different groups η and λ by considering similar reasoning mentioned before for $[\mathbf{D_g}]_{\gamma,\zeta}$, and the entry located at the main diagonal of this matrix is equal to the average degree of coherency between the generators of the corresponding group. Therefore, if the system separation set, i.e. **J**, has the following definition, its each member assesses the weakness of coupling between two coherent groups.

$$\mathbf{J} = \left\{ \frac{[\mathbf{D_c}]_{\eta,\eta} + [\mathbf{D_c}]_{\lambda,\lambda}}{2 \times [\mathbf{D_c}]_{\eta,\lambda}} \;\middle|\; \eta = 1, 2, ..., N_c \text{and} \lambda = \eta, \eta + 1, ..., N_c \right\} \tag{4}$$

Because when the study system operates normally, all of its generators form a group within every time window, and thus 1 is the only member of **J**; however, after occurring a disturbance and forming different coherent groups, the value of the corresponding member of two groups first increases at the consequent time windows that means strengthening of the coupling between the generators of each group and weakening of the coupling between the groups. By passing time, the coupling between the groups may become so weak that they arrive on the verge of decoupling. This situation relates to a sharp decrease in the value of the corresponding member that turns this value into a negative number in the specified time window. So, the smallest member of **J** can be considered for defining *SSI* to determine the time window of happening the aforementioned situation for the first pair of coherent groups. It is to be noted that the data required for the calculation of *SSI* is continuously updated and the procedure of the previous and this section is repeated in the sliding time window moved every T_q second until a change in the sign of *SSI* is observed in a time window. The end of this time window, i.e. T_h, can be regarded as the proper time to announce the system need for intentional islanding. It is worth citing that T_q must be greater than the calculation time of each iteration. Although this time depends on the size of the study system and the processing speed of the utilized computer, but it can be constant, if the faster data acquisition and analyzing tools and stronger computers are used in larger systems to detect separation state of coherent generator groups as an alarm to start designing the islanding scheme.

3. Location of islanding

3.1. Finding initial islands

After obtaining T_h, the method proposed in Ref. [34] for tracing active power productions of the generators is used to determine the amounts of active power consumptions of the loads supplied by each generator and shares of the generators in active power flows of the lines for acquiring initial allocation of non-generation buses to the island of each coherent group by the following rules:

Rule 1: If all active power consumption of each load bus is supplied by the generators of a coherent group, put this bus into the island of that group.

Rule 2: If all active power flows of all incident lines of each transmission bus, on which there are no generators and loads, are provided by the generators of a coherent group, put this bus into the island of that group.

It is worth noting that all data needed to implement the method of Ref. [34], i.e. the values of active power productions and consumptions of the buses and the amounts of active power flows of the lines at $t = T_h$, can be attained based on the variables measured by WAMS.

3.2. Finding final islands

After the initial assignment of buses to the islands, some buses may remain unallocated. These ones named outlier buses constitute $\mathbf{O_b}$ set. Furthermore, a line that connects to at least one outlier bus is called outlier line and belongs to $\mathbf{O_l}$ set.

To acquire final islands, the allocation of outlier buses is formulated as a BLP problem. To do so, the binary variables, e.g. $x_{\alpha,\eta}$, the number of which equals N_c, are first defined for the status of each outlier bus as follows:

If outlier bus α is located within island η, i.e. it belongs to $\mathbf{I}_\eta$ set, $x_{\alpha,\eta}$ is equal to one, otherwise $x_{\alpha,\eta} = 0$.

Hence, the following constraint must be regarded to ensure that each outlier bus locates within only one island:

$$\sum_{\eta=1}^{N_c} x_{\alpha,\eta} = 1 \quad \text{for} \alpha \in \mathbf{O_b} \tag{5}$$

As shown below, one or N_c binary variables, e.g. $y_{\alpha-\beta,\eta}$, are then considered for the status of each outlier line depending on whether one or both of its end buses are outlier bus to model the rule that each outlier line is closed, i.e. $y_{\alpha-\beta,\eta} = 1$, only if its end buses are located within the same island, i.e. α and $\beta \in \mathbf{I}_\eta$:

- If bus α is outlier bus and bus β is located within island η, then the status of the line between buses α and β, i.e. $y_{\alpha-\beta,\eta}$, is equal to $x_{\alpha,\eta}$.
- If buses α and β belong to $\mathbf{I}_\eta$ and $\mathbf{O_b}$, respectively, then $y_{\alpha-\beta,\eta} = x_{\beta,\eta}$.
- If α and $\beta \in \mathbf{O_b}$, then $y_{\alpha-\beta,\eta} \leq x_{\alpha,\eta}$ & $y_{\alpha-\beta,\eta} \leq x_{\beta,\eta}$ & $y_{\alpha-\beta,\eta} \geq x_{\alpha,\eta} + x_{\beta,\eta} - 1$ for $\eta = 1, 2, \ldots, N_c$.

Finally, to disconnect the lines that transmit the smallest apparent power, the following objective function is used for this problem:

$$\text{Minimize} - \left(\sum_{\text{Line}_{\alpha-\beta} \in \mathbf{O_l}} \sum_{\eta=1}^{N_c} S_{\text{Line}_{\alpha-\beta}} \times y_{\alpha-\beta,\eta} \right) \tag{6}$$

where $\text{Line}_{\alpha-\beta}$ is a symbol of the line between buses α and β and $S_{\text{Line}_{\alpha-\beta}}$ is the maximum amount of apparent power flows at both ends of $\text{Line}_{\alpha-\beta}$ in MVA for $t = T_h$.

Solving this problem determines each outlier bus is the member of which island. Consequently, final islands can be acquired by disconnecting the lines whose end buses belong to two different islands. The flowchart illustrated in Fig. 1 summarizes the procedures discussed in Sections 2 and 3.

4. Simulation results

The proposed strategy is implemented on the IEEE 39- and 118-bus test systems, using MATLAB, to evaluate its effectiveness. The data needed for simulating dynamic behavior of these 60^{Hz} systems is taken from Refs. [42] and [10], respectively. Figs. 2 and 3 display the graphs of these systems. In these figures, numbers of the generator and non-generation buses are depicted in solid-squares by the light and dark colors, respectively.

In the proposed approach, it is considered that $T_w = 1^{\text{sec}}$, $T_q = 0.1^{\text{sec}}$ for 39-bus system, and $T_q = 0.2^{\text{sec}}$ for 118-bus system on a PC with Intel Pentium 3.00-GHz processor and 4-GB of RAM. Moreover, four following criteria are employed for comparison of different islanding schemes:

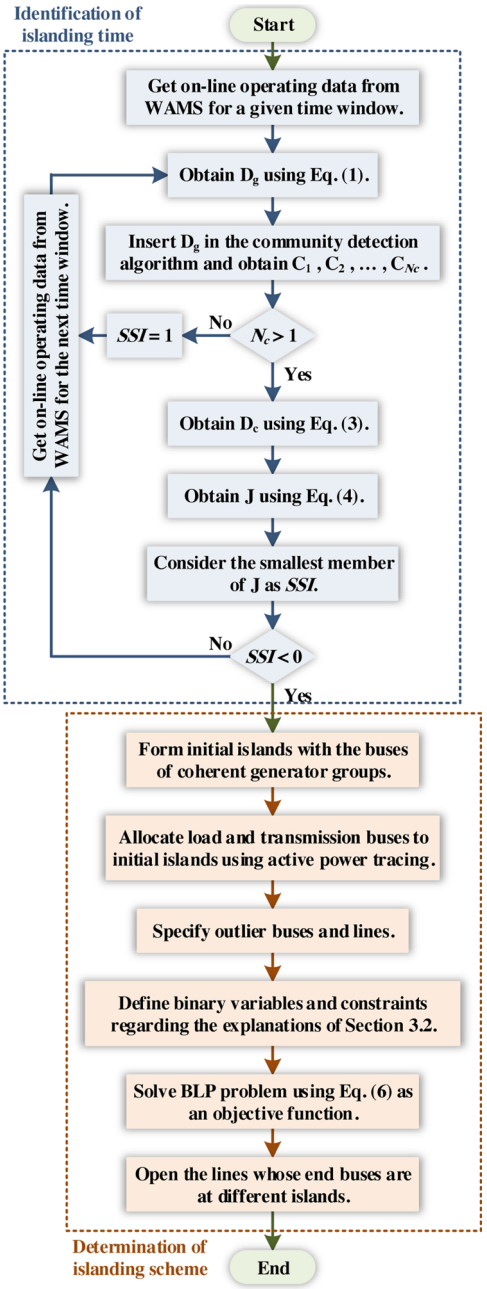


Fig. 1. Flowchart of the proposed approach for controlled splitting.

1) S_Γ: Total amount of apparent power flows of the disconnected lines, which create Γ set, in MVA for $t = T_h$.
2) ΔP_{max}: Maximum generation-demand imbalance of islands in % for $t = T_h$.
3) ΔV_{max}: Maximum voltage deviation of buses from 1 p.u.
4) Δf_{max} : Maximum frequency deviation of islands from 100%

In continuation of this section, we describe the simulation details and discuss the results.

4.1. IEEE 39-bus test system

In this system, first, a three phase short-circuit occurs on bus 29 at $t = 1.69^{sec}$. This fault is cleared after 0.15^{sec} by removing $Line_{28-29}$. Then, another three-phase short-circuit happens on bus 4 at $t = 1.85^{sec}$, which is cleared after 0.15^{sec} by opening $Line_{4-5}$ [5]. Fig. 4 shows the voltage magnitudes of the buses in the case of these disturbances and

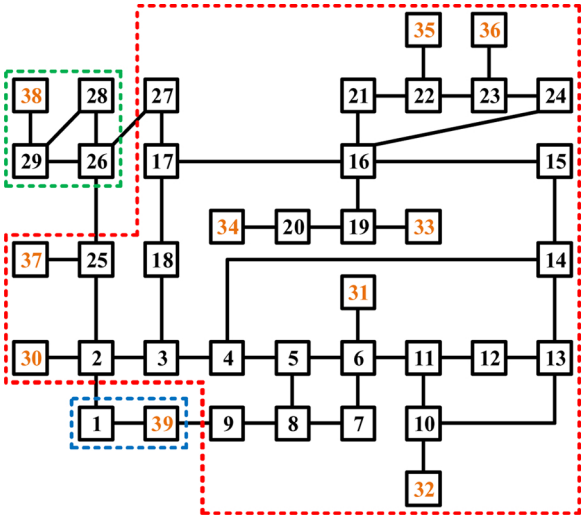


Fig. 2. Graph of the IEEE 39-bus system.

without controlled splitting, which clearly reflects the system instability.

SSI for this case is given in Table 1. Based on this table, *SSI* is positive in the time window of (1.1^{sec}, 2.1^{sec}) and is negative in the time window of (1.2^{sec}, 2.2^{sec}). Because, $\mathbf{D_g}$ can be written as follows in the later time window:

$$\mathbf{D_g} = \begin{bmatrix} 1 & 0.83 & 0.85 & 0.87 & 0.95 & 0.91 & 0.86 & 0.91 & 0.06 & 0.76 \\ 0.83 & 1 & 0.98 & 0.94 & 0.88 & 0.92 & 0.96 & 0.87 & 0.22 & 0.59 \\ 0.85 & 0.98 & 1 & 0.96 & 0.89 & 0.94 & 0.98 & 0.89 & 0.21 & 0.61 \\ 0.87 & 0.94 & 0.96 & 1 & 0.94 & 0.97 & 0.98 & 0.95 & 0.18 & 0.64 \\ 0.95 & 0.88 & 0.89 & 0.94 & 1 & 0.95 & 0.93 & 0.97 & 0.13 & 0.72 \\ 0.91 & 0.92 & 0.94 & 0.97 & 0.95 & 1 & 0.95 & 0.94 & 0.15 & 0.67 \\ 0.86 & 0.96 & 0.98 & 0.98 & 0.93 & 0.95 & 1 & 0.93 & 0.20 & 0.62 \\ 0.91 & 0.87 & 0.89 & 0.95 & 0.97 & 0.94 & 0.93 & 1 & 0.14 & 0.70 \\ 0.06 & 0.22 & 0.21 & 0.18 & 0.13 & 0.15 & 0.20 & 0.14 & 1 & -0.17 \\ 0.76 & 0.59 & 0.61 & 0.64 & 0.72 & 0.67 & 0.62 & 0.70 & -0.17 & 1 \end{bmatrix} \quad (7)$$

By inserting this matrix in the community detection algorithm, as conformed to Fig. 5, this algorithm constructs one group from the generator on bus 38, i.e. G_{38}, which is the ninth generator of the system, puts the generator on bus 39, i.e. G_{39}, which is the tenth generator of the system, into the other group, and constitutes the third group from the rest of the generators. This means that $N_c = 3$ and $\mathbf{C}_1$, $\mathbf{C}_2$ and $\mathbf{C}_3$ will take the following forms:

$$\begin{aligned} \mathbf{C}_1 &= \{1, 2, 3, 4, 5, 6, 7, 8\} \\ \mathbf{C}_2 &= \{9\} \\ \mathbf{C}_3 &= \{10\} \end{aligned} \quad (8)$$

Using Eqs. (2) and (3), $\mathbf{D_c}$ can be attained as:

$$\mathbf{D_c} = \begin{bmatrix} 0.92 & 0.15 & 0.67 \\ 0.15 & 1 & -0.17 \\ 0.67 & -0.17 & 1 \end{bmatrix} \quad (9)$$

And thus $\mathbf{J}$ can be acquired via applying Eq. (4) as follows:

$$\mathbf{J} = \{1, 6.21, 1.44, -5.81\} \quad (10)$$

The smallest member of $\mathbf{J}$ shows the value of *SSI*, i.e. -5.81. Therefore, designing the islanding scheme must be initiated at $T_h = 2.20^{sec}$ and three islands must be formed. The buses of initial islands, which is determined by tracing active power productions of the generators, are reported in Table 2. According to this table and taking Fig. 2 into account, since bus 9 is located within island 1 and bus 39 belongs to island 3, $Line_{9-39}$ must be opened. On the other hand, $\mathbf{O_b}$ and $\mathbf{O_l}$ can be specified as:

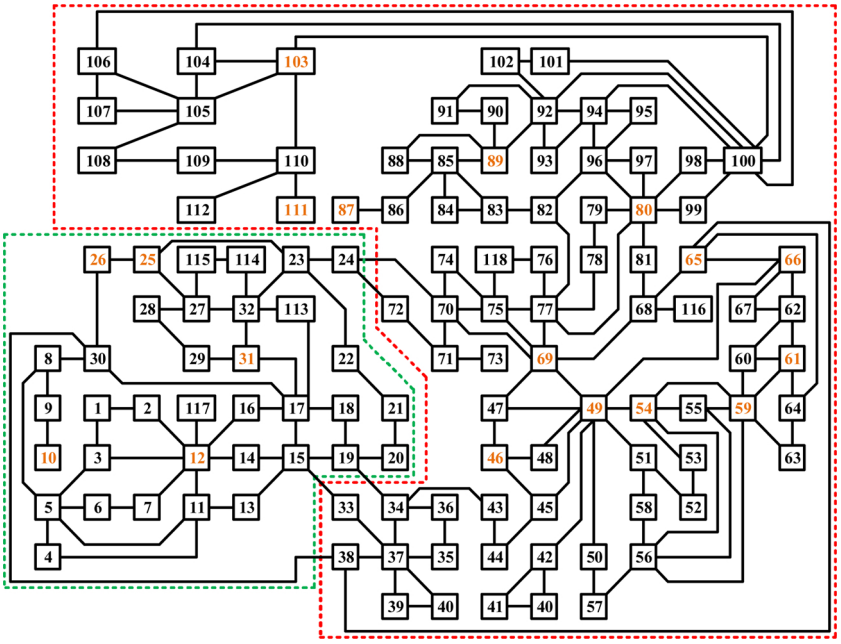


Fig. 3. Graph of the IEEE 118-bus system.

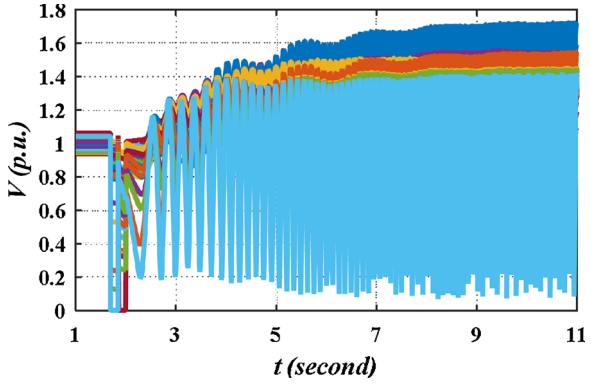


Fig. 4. Voltage magnitudes of buses without intentional islanding of the IEEE 39-bus system.

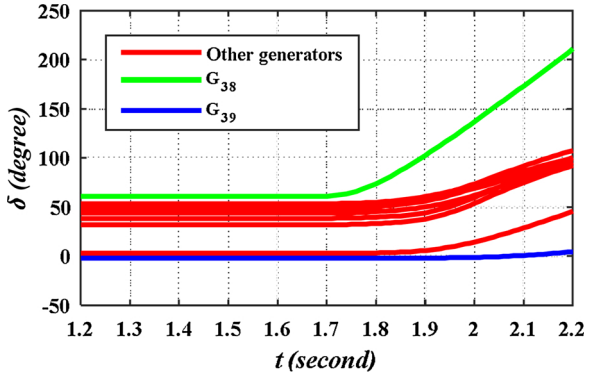


Fig. 5. Generator rotor angles of the IEEE 39-bus system for the time window in which $SSI < 0$.

$$\mathbf{O_b} = \{1, 2, 25, 27\} \tag{11}$$

$$\mathbf{O_l} = \{\text{Line}_{1-2}, \text{Line}_{1-39}, \text{Line}_{2-3}, \text{Line}_{2-25}, \text{Line}_{2-30}, \text{Line}_{17-27}, \text{Line}_{25-26}, \text{Line}_{25-37}, \text{Line}_{26-27}\} \tag{12}$$

Hence, with attention to the explanations presented in Section 3.2, the following constraints must be regarded for the BLP problem:

$$\begin{aligned} &1 \in \mathbf{O_b} \xrightarrow{N_c=3} x_{1,1} + x_{1,2} + x_{1,3} = 1 \\ &2 \in \mathbf{O_b} \xrightarrow{N_c=3} x_{2,1} + x_{2,2} + x_{2,3} = 1 \\ &25 \in \mathbf{O_b} \xrightarrow{N_c=3} x_{25,1} + x_{25,2} + x_{25,3} = 1 \\ &27 \in \mathbf{O_b} \xrightarrow{N_c=3} x_{27,1} + x_{27,2} + x_{27,3} = 1 \end{aligned} \tag{13}$$

$$(\text{Line}_{1-2} \in \mathbf{O_l}) \wedge (1\text{and}2 \in \mathbf{O_b}) \rightarrow \begin{cases} y_{1-2,1} \le x_{1,1} \\ y_{1-2,1} \le x_{2,1} \\ y_{1-2,1} \ge x_{1,1} + x_{2,1} - 1 \end{cases}, \begin{cases} y_{1-2,2} \le x_{1,2} \\ y_{1-2,2} \le x_{2,2} \\ y_{1-2,2} \ge x_{1,2} + x_{2,2} - 1 \end{cases}, \begin{cases} y_{1-2,3} \le x_{1,3} \\ y_{1-2,3} \le x_{2,3} \\ y_{1-2,3} \ge x_{1,3} + x_{2,3} - 1 \end{cases} \tag{14}$$

Table 2
Buses of initial islands for the IEEE 39-bus system.

Island	Generator bus	Load bus	Transmission bus
$\mathbf{I}_1$	30, 31, 32, 33, 34, 35, 36, 37	3, 4, 7, 8, 12, 15, 16, 18, 20, 21, 23, 24	5, 6, 9, 10, 11, 13, 14, 17, 19, 22
$\mathbf{I}_2$	38	26, 28, 29	–
$\mathbf{I}_3$	39	–	–

$$(\text{Line}_{1-39} \in \mathbf{O_l}) \wedge (1 \in \mathbf{O_b}) \wedge (39 \in \mathbf{I}_3) \rightarrow y_{1-39,3} = x_{1,3} \tag{15}$$

$$(\text{Line}_{2-3} \in \mathbf{O_l}) \wedge (2 \in \mathbf{O_b}) \wedge (3 \in \mathbf{I}_1) \rightarrow y_{2-3,1} = x_{2,1} \tag{16}$$

$$(\text{Line}_{2-25} \in \mathbf{O_l}) \wedge (2\text{and}25 \in \mathbf{O_b}) \rightarrow \begin{cases} y_{2-25,1} \le x_{2,1} \\ y_{2-25,1} \le x_{25,1} \\ y_{2-25,1} \ge x_{2,1} + x_{25,1} - 1 \end{cases}, \begin{cases} y_{2-25,2} \le x_{2,2} \\ y_{2-25,2} \le x_{25,2} \\ y_{2-25,2} \ge x_{2,2} + x_{25,2} - 1 \end{cases}, \begin{cases} y_{2-25,3} \le x_{2,3} \\ y_{2-25,3} \le x_{25,3} \\ y_{2-25,3} \ge x_{2,3} + x_{25,3} - 1 \end{cases} \tag{17}$$

Table 1
System separation indicator of the IEEE 39-bus system without intentional islanding.

Time window (sec)	(0.0, 1.0) to (0.7, 1.7)	(0.8, 1.8)	(0.9, 1.9)	(1.0, 2.0)	(1.1, 2.1)	**(1.2, 2.2)**
SSI	1.00	0.99	0.95	1.76	1.26	**−5.81**

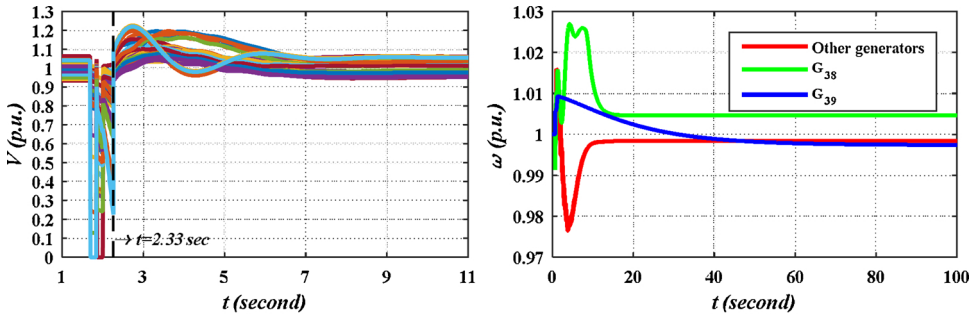


Fig. 6. With intentional islanding of the IEEE 39-bus system.
(a) Voltage magnitudes of buses.
(b) Angular velocities of generators.

Table 3
Results for intentional islanding of the IEEE 39-bus system.

Method	N_c	size(Γ)	S_Γ (MVA)	ΔP_{max} (%)	ΔV_{max} (p.u.)	Δf_{max} (%)
Proposed method	3	4	835.00	22.25	0.064	0.488
Ref. [5]	4	6	945.27	57.78	0.223	2.300

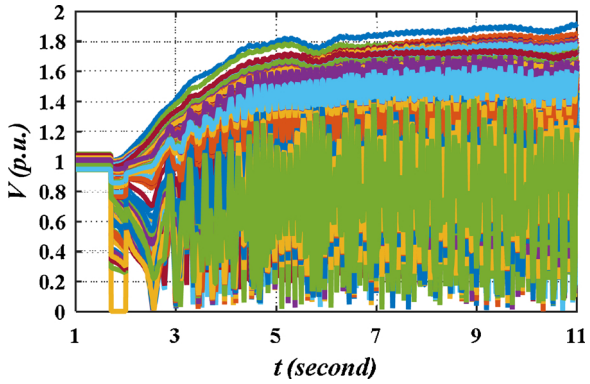


Fig. 7. Voltage magnitudes of buses without intentional islanding of the IEEE 118-bus system.

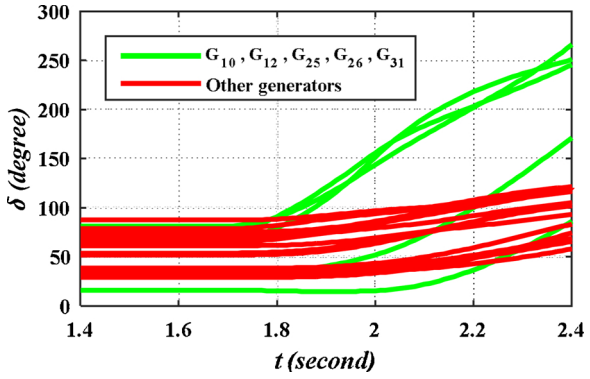


Fig. 8. Generator rotor angles of the IEEE 118-bus system for the time window in which $SSI < 0$.

$$(\text{Line}_{2-30} \in \mathbf{O_l}) \wedge (2 \in \mathbf{O_b}) \wedge (30 \in \mathbf{I_l}) \rightarrow y_{2-30,1} = x_{2,1} \tag{18}$$

$$(\text{Line}_{17-27} \in \mathbf{O_l}) \wedge (17 \in \mathbf{I_l}) \wedge (27 \in \mathbf{O_b}) \rightarrow y_{17-27,1} = x_{27,1} \tag{19}$$

$$(\text{Line}_{25-26} \in \mathbf{O_l}) \wedge (25 \in \mathbf{O_b}) \wedge (26 \in \mathbf{I_2}) \rightarrow y_{25-26,2} = x_{25,2} \tag{20}$$

$$(\text{Line}_{25-37} \in \mathbf{O_l}) \wedge (25 \in \mathbf{O_b}) \wedge (37 \in \mathbf{I_l}) \rightarrow y_{25-37,1} = x_{25,1} \tag{21}$$

$$(\text{Line}_{26-27} \in \mathbf{O_l}) \wedge (26 \in \mathbf{I_2}) \wedge (27 \in \mathbf{O_b}) \rightarrow y_{26-27,2} = x_{27,2} \tag{22}$$

And the following solution can be achieved using Eq. (6):

$$\left.\begin{matrix} x_{1,1}=0 \\ x_{1,2}=0 \\ x_{1,3}=1 \end{matrix}\right\} \rightarrow 1 \in \mathbf{I}_3 \quad , \quad \left.\begin{matrix} x_{2,1}=1 \\ x_{2,2}=0 \\ x_{2,3}=0 \end{matrix}\right\} \rightarrow 2 \in \mathbf{I}_1$$
$$\left.\begin{matrix} x_{25,1}=1 \\ x_{25,2}=0 \\ x_{25,3}=0 \end{matrix}\right\} \rightarrow 25 \in \mathbf{I}_1 \quad , \quad \left.\begin{matrix} x_{27,1}=1 \\ x_{27,2}=0 \\ x_{27,3}=0 \end{matrix}\right\} \rightarrow 27 \in \mathbf{I}_1 \tag{23}$$

$$(y_{1-2,1}=0) \vee (y_{1-2,2}=0) \vee (y_{1-2,3}=0) \rightarrow \text{Line}_{1-2}\text{is open}$$
$$y_{1-39,3}=1 \rightarrow \text{Line}_{1-39}\text{is closed}$$
$$y_{2-3,1}=1 \rightarrow \text{Line}_{2-3}\text{is closed}$$
$$(y_{2-25,1}=1) \vee (y_{2-25,2}=0) \vee (y_{2-25,3}=0) \rightarrow \text{Line}_{2-25}\text{is closed}$$
$$y_{2-30,1}=1 \rightarrow \text{Line}_{2-30}\text{is closed}$$
$$y_{17-27,1}=1 \rightarrow \text{Line}_{17-27}\text{is closed}$$
$$y_{25-26,2}=0 \rightarrow \text{Line}_{25-26}\text{is open}$$
$$y_{25-37,1}=1 \rightarrow \text{Line}_{25-37}\text{is closed}$$
$$y_{26-27,2}=0 \rightarrow \text{Line}_{26-27}\text{is open} \tag{24}$$

In other words,

$$\Gamma = \{\text{Line}_{1-2}\,,\ \text{Line}_{25-26}\,,\ \text{Line}_{26-27}\,,\ \text{Line}_{9-39}\} \tag{25}$$

Final islands are illustrated in Fig. 2 by dashed lines. Fig. 6 demonstrates that the disconnection of Line_{1-2}, Line_{25-26}, Line_{26-27}, and Line_{9-39} at $t = 2.33^{\text{sec}}$ (the sum of T_h plus 0.03^{sec} for performing calculations on the aforesaid PC plus 2 cycles for relays' time plus 2 cycles

Table 4
System separation indicator of the IEEE 118-bus system without intentional islanding.

Time window (sec)	(0.0, 1.0) to (0.6, 1.6)	(0.8, 1.8)	(1.0, 2.0)	(1.2, 2.2)	**(1.4, 2.4)**
SSI	1.00	0.99	1.45	8.24	**−48.15**

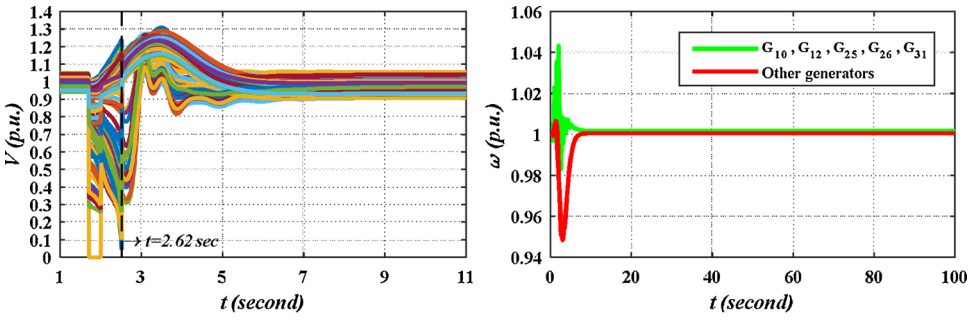


Fig. 9. With intentional islanding of the IEEE 118-bus system.
(a) Voltage magnitudes of buses.
(b) Angular velocities of generators.

Table 5
Results for intentional islanding of the IEEE 118-bus system.

Method	N_c	size(Γ)	S_Γ (MVA)	ΔP_{max} (%)	ΔV_{max} (p.u.)	Δf_{max} (%)
Proposed method	2	5	147.53	7.51	0.095	0.180
Ref. [6]	2	6	310.77	10.74	0.138	0.551
Ref. [9]	2	6	204.24	10.08	0.180	0.530
Ref. [11]	2	6	329.84	11.34	0.170	0.636
Ref. [12,13]	3	9	234.44	30.48	0.095	0.806
Ref. [14]	2	6	328.87	11.78	0.180	0.710
Ref. [15]	2	6	323.64	9.85	0.159	0.456
Ref. [16]	3	10	286.86	36.80	0.106	1.049
Ref. [22]	2	7	188.50	8.12	0.201	0.286
Ref. [25]	3	10	277.89	34.54	0.106	0.880

for practical minimum communication time delays plus 2 cycles for breakers' opening time [43,44]) results in acceptable voltage magnitudes for the buses and angular velocities for the generators ultimately, and makes the system stable.

In Table 3, the proposed scheme is compared with the scheme presented in Ref. [5]. This table shows that the proposed method has achieved the best cut set; because the obtained cut set consists of the lines that transmit the lowest apparent power at T_h, and considering the values of active power productions and consumptions at this time, disconnecting four lines of this set forms the islands with minimum generation-demand imbalance percentage. This scheme is also better for maintaining the grid integrity as it splits the system into smaller number of islands. Furthermore, the proposed scheme achieves the smallest deviation for the voltage magnitudes of the buses and the frequencies of the islands in the steady state.

4.2. IEEE 118-bus test system

In this system, the fault is a three phase short-circuit on Line_{23-25} and located very close to bus 25 at $t = 1.70^{\text{sec}}$. This fault is cleared after 0.30^{sec} by removing the faulted line [6,9,11–16,22,25]. The voltage magnitudes of the buses in the case of this disturbance and without controlled splitting are depicted in Fig. 7, which obviously shows the system instability.

According to Table 4, which indicates *SSI* of this system, the time of designing the islanding scheme is $T_h = 2.40^{\text{sec}}$. As seen in Fig. 8, the proposed method identifies two coherent groups of the generators in this case, one group consists of $\{G_{10}, G_{12}, G_{25}, G_{26}, G_{31}\}$ and another group includes the other generators. To put these groups into their respective islands, the proposed method suggests disconnecting Line_{15-33}, Line_{19-34}, Line_{24-70}, Line_{24-72} and Line_{30-38} as illustrated in Fig. 3 by dashed lines. The execution of this scheme at $t = 2.62^{\text{sec}}$ (the sum of T_h plus 0.12^{sec} for accomplishing calculations on the aforementioned PC plus 2 cycles for relays' time plus 2 cycles for practical minimum communication time delays plus 2 cycles for breakers' opening time) will stabilize the system by reaching the buses' voltage magnitudes and the generators' angular velocities to acceptable values as observed in Fig. 9.

Comparison of the proposed scheme with the schemes of the references mentioned in Table 5 demonstrates its superiority because it has achieved the least value for size(Γ), S_Γ, ΔP_{max}, ΔV_{max}, and Δf_{max}.

5. Conclusion

In this paper, an approach was presented to inhibit the propagation of disturbances and system collapse through intentional islanding. In this approach, determining the cut set to form stable islands comprises of two stages: the identification of coherent generator groups, each of them locates within an island, and the allocation of non-generation buses to the islands. In the first stage, a weighted undirected graph, whose vertices correspond to the system generators, was considered. The weight of each edge was then defined based on the degree of the similarity of rotor angles of the generators at its both sides in the time window of interest. Afterward, the adjacency matrix of this graph was fed into an existing community detection algorithm. The value of each component of the output vector of this algorithm represented the group number of its corresponding vertex. Therefore, the largest component signified the total number of groups, and it was not necessary to predetermine this number as an algorithm input. In the second stage, the previous authors work in Ref. [34] for tracing active power productions of the generators was used to form an initial islands by assigning the load and transmission buses, whose entire active power consumptions and flows of their all incident lines respectively was supplied by a coherent group of the generators, to the domain of that group. Subsequently, a BLP problem was solved to allocate the buses, which did not belong to any initial island, to these islands such that the total apparent power transmitted by the lines, whose end buses were at different islands, would be minimal. In addition, an indicator was defined so that becoming its value less than a threshold would represent system instability and the requirement for intentional islanding. The key advantage of the proposed indicator was that its threshold was equal to zero in all systems and operating conditions. The controlled splitting of two standard systems by the proposed approach revealed that it is more efficient as compared to some prior methods.

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